



Linear Models



Data Analytics Lab



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Where Linear Models Fit

Classification methods can be mapped into families: tree-based, probabilistic, distance-based, margin-based, ensemble — and linear models. A linear model predicts class membership through a weighted sum of the input attributes, so the decision boundary is a hyperplane in feature space.

Simplicity as a Feature

Simplicity, not a limitation. A linear boundary is easy to inspect, easy to explain to a non-technical stakeholder, and hard to overfit on small datasets.

The Natural First Alternative

The natural first alternative to a tree. Where `cla_dtree` splits the space into axis-aligned regions, a GLM draws one smooth separating surface.

Foundation for Regularization

Foundation for everything regularized. L1/L2 penalties, elastic nets, and even the linear layer inside a neural network all extend the same idea.

Logistic Regression – `cla_glm`

Binary logistic regression through R's Generalized Linear Model framework. The target must be binary; the log-odds of the positive class are modeled as a linear combination of the predictors.

```
model <- cla_glm("IsVersicolor",  
                positive = "versicolor")  
model <- fit(model, iris_train)  
prediction <- predict(model, iris_test)  
evaluate(model,  
          iris_test$IsVersicolor,  
          prediction)
```

- **Binary target required:** the `daltoolbox` example remaps the three-class `iris$Species` into `versicolor` vs. `not_versicolor`.
- **Same Experiment Line:** `cla_glm(target, positive = ...) → fit() → predict() → evaluate()`.
- **Interpretable coefficients:** directly readable as log-odds contributions — a rare property among this collection's classifiers.

Reference: McCullagh, P., and Nelder, J. A. (1989). *Generalized Linear Models*.

L1-Regularized Logistic Regression – `cla_glmnet`

Keeps the exact same binary classification line of experiment as `cla_glm`, but replaces ordinary maximum-likelihood fitting with L1-penalized (lasso) fitting, which can shrink irrelevant coefficients to exactly zero.

```
model <- cla_glmnet("IsVersicolor",  
  lambda = "lambda.1se")  
model <- fit(model, iris_train)
```

- **One extra knob:** `lambda` controls how aggressively coefficients are penalized.
- **Conservative default:** `lambda = "lambda.1se"` picks the most regularized model within one standard error of the best cross-validated fit.
- **When to use it:** the predictor set is large or collinear and a plain GLM would overfit or become numerically unstable.

Reference: Friedman, J., Hastie, T., and Tibshirani, R. (2010). Regularization Paths for GLMs via Coordinate Descent.

What to Observe

- **Same experiment body:** both examples keep the exact same experiment body used across the whole classification collection — only the learner and the target's binary requirement change.
- **Different strengths:** `cla_glm` favors interpretability; `cla_glmnet` favors robustness under many or correlated predictors.
- **The linear baseline:** the linear counterpart to every non-linear method covered elsewhere (kNN, SVM, trees, ensembles, neural networks).



Acknowledgements



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